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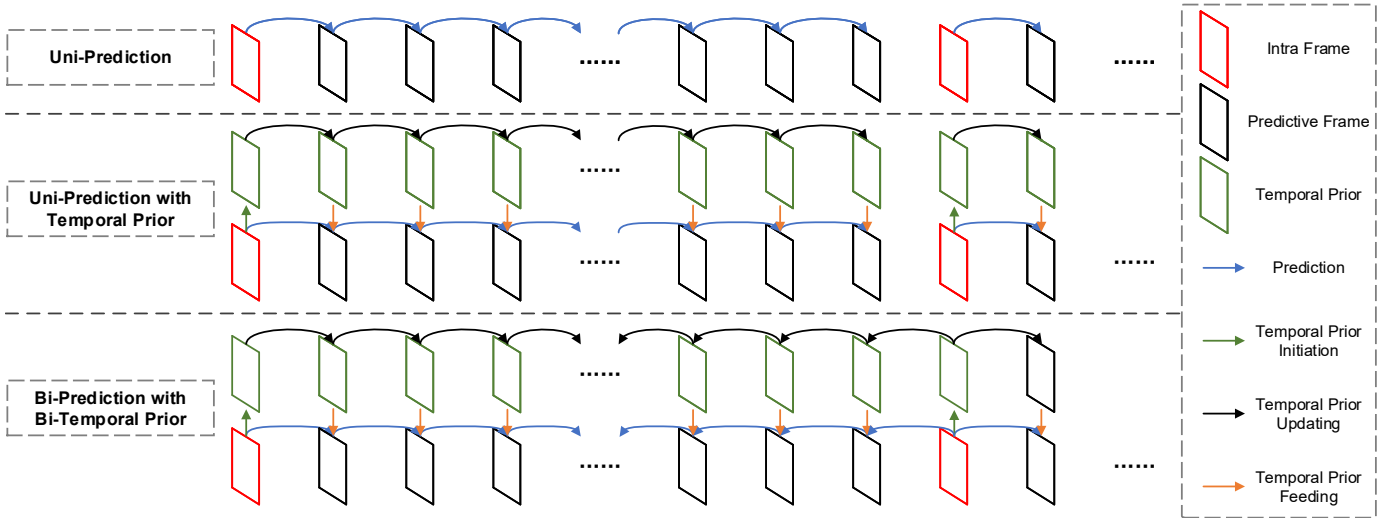


Fig. 2: The inference paradigm difference between unidirectional prediction (Uni-Prediction) and bidirectional prediction (Bi-Prediction). Most existing video compression methods adopt Uni-Prediction scheme since it is simple and easy to deploy. The baseline method LSTVC enhance the Uni-Prediction with temporal prior. In this paper, we propose Bi-Prediction scheme and bidirectional temporal prior to alleviate error propagation.

robust yet efficient inter coding. Progressive guided motion compensation uses a hierarchical structure to achieve multi-scale compensation, and its smaller-scale compensation results will be used to guide the offset prediction of the next scale, improving the stability of the entire compensation process. Moreover, we extend the framework with bidirectional prediction and propose bidirectional temporal prior to further alleviate error propagation. We conduct experiments towards practical scenarios by setting the intra period to 32, and the effectiveness of BiTVC is clearly shown, as it achieves competitive rate-distortion performance with the standard reference software HM-16.22 while maintaining high efficiency.

II. PROPOSED METHOD

In this section, we first describe the inter-frame coding, including the temporal prior and the progressive guided motion compensation. Then we introduce the proposed framework and the overall inference process. Finally, the training scheme and loss functions are present.

A. Inter-frame Coding

We use LSTVC[6] as our baseline method, and Figure-1 provides a high-level overview of our inter-frame compression model. As the conditional coding-based method[9], we abnegate the residue coding-based paradigm and use the contextual information to compress details. First, we estimate the motion vectors between the reference frame and the current frame, then the motion compression module transforms the motion vectors into the latent representation. Reconstructed motion vectors from the motion decoder will be sent to accomplish progressive guided motion compensation with the reference frame. The compensation results are regarded as context to facilitate contextual compression, including contextual encoder, decoder, and entropy model optimization.

The temporal information is extracted from each decoded frame and supplemented to the temporal prior. Besides, the temporal prior will be updated explicitly with the decoded motion vectors during compression and should be encoded into the latent representation before participating in the entropy model. For contextual compression, we follow LSTVC and use long- and short-range temporal information simultaneously to facilitate distribution parameter prediction.

1) *Temporal Prior*: Priors of the entropy model such as hyperprior[4] and hierarchical context model[5] always play a significant role in predicting more accurate parameters of Gaussian distribution and enhancing rate-distortion performance. Valuable prior information can greatly reduce the Shannon cross entropy between the predicted distribution of the latent representation and the actual marginal distribution. Therefore, many video compression methods[1], [2], [3] directly follow the image compression work and adopt hyperprior or autoregressive context model for distribution parameter prediction for compression.

We adopt the prior updating and usage scheme from LSTVC. For initiation, we generate the temporal prior for each GOP by extracting prior features from the I frame before compressing the first P frame. Besides, the temporal prior is updated under two conditions. First, to generate a temporal prior with a higher spatial correlation to the current frame, we align and update the temporal prior with the decoded motion vector before contextual compression. Second, except for the first P frame, we update the prior with the decoded frame contents before compressing the current motion vectors. The updated temporal prior can significantly reduce spatial statistical dependencies. For the motion and contextual entropy model, we use the prior encoders to transform the temporal prior for the corresponding entropy model.

2) *Progressive Guided Motion Compensation*: Without prompt constrain and guidance, kernel offsets of deformable convolution are difficult to estimate accurately, and the large range of motion may cause offset prediction overflow. In this paper, we use the progressive guided motion compensation (PGMC) from LSTVC which uses the core idea of optical flow guiding convolution and scale-by-scale guided reconstruction. Besides, we adopt the first-order alignment and use optical flow to guide the generation of the kernel offsets for achieving stable training and accurate compensation.

To avoid overflow of the offsets prediction, PGMC prioritizes motion compensation at the smaller-scale level and guide offset prediction and feature reconstruction for the next scale, which means smaller-scale compensation aims at coarse offset prediction and frame reconstruction. The offsets generated by guidance participate in motion compensation, and the compensation results continue to guide the offsets prediction for the next scale, forming a progressively guided compensation scheme.

B. Framework Description

In this paper, to handle the error propagation in the low-latency scenarios, we extend the baseline framework to the bidirectional prediction mode. Figure 2 shows the paradigm difference between the unidirectional prediction (Uni-prediction) mode and our proposed bidirectional prediction mode. Most previous learned video compression methods[1], [2], [3], [9] use Uni-prediction strategy to accomplish serial frame prediction. Uni-prediction is easy to implement and the prediction order matches low-latency scenarios, however, it suffers from severe error propagation. Temporal prior can mitigate the magnitude of error propagation in Uni-prediction since it contains global temporal information to assist compression, but it cannot tackle the problem effectively.

We extend the framework with the bidirectional prediction (Bi-prediction) and propose bidirectional temporal prior (Bi temporal prior) to boost compression performance. Typically, the reconstruction quality of the I frame is much higher than the quality of the subsequent P frames, and a reference frame with higher quality will facilitate the reconstruction of the next frame due to its richer texture and more accurate motion information. So Bi-prediction can provide higher-quality reference frames from two I frame, and the intermediate predictive frames are predicted from different I frames by forward and backward prediction directions. Furthermore, combined with the Bi-prediction scheme, we propose Bi temporal prior to provide more accurate mutual information, which is also updated with the bidirectional inference.

C. Loss Function

We adopt multi-stage training strategy and train motion-related modules at first to further improve spatial relevance between the context prior and the current frame. Specifically, our framework focus on stabilizing compensation performance

at first and only motion-related modules are engaged in training. The rate-compensation loss L_{r-c} is used in this stage:

$$L_{r-c} = \lambda \cdot D_{com} + R = \lambda \cdot d(\mathbf{cxt}, F_t) + R_{\tilde{m}}, \quad (1)$$

in which $\mathbf{cxt}$ denotes the compensation results, F_t is the feature extracted from the current frame, and λ is the Lagrange multiplier that determines the trade-off between the predicted entropy results R and the compensation distortion D_{com} .

After the convergence of the motion related module, we use the total RD optimization loss function:

$$L_{total} = \lambda \cdot D + R = \lambda \cdot d(\hat{I}_t, I_t) + R_{\tilde{m}} + R_{\tilde{c}} \quad (2)$$

where $d(I_t, \hat{I}_t)$ denotes the distortion between I_t and $\hat{I}_t$. $R_{\tilde{m}}$ and $R_{\tilde{c}}$ stand for the estimated rate for motion and contextual compression. We use MSE as the distortion function for both intra and inter compression model.

III. EXPERIMENTS

A. Datasets and Implementation Details

1) *Datasets*: We use the Flicker2W and the Vimeo90K dataset as the training dataset for intra and inter compression training, respectively. In Vimeo90K, each video sample contains seven consecutive frames. To validate the effectiveness of our model and compare performance with other video compression approaches, we test our models on the challenge validation datasets, and the resolution of this dataset is 1920×1080 .

2) *Training settings*: During training, our methods require the previous reconstructed frame $\hat{I}_{t-1}$ for motion compensation, so we train our model six times in succession using seven consecutive frames by buffering the previous reconstructed frame. With regard to Intra frame compression, we choose TinyLIC[10] to achieve higher rate-distortion performance than BPG[11] while maintaining efficiency.

We train nine intra-frame coding models and train six inter-frame coding models with different λ to control RD performance. The batch size is set to 16, and we randomly crop the training sequences into the resolution of 256×256 . The AdamW optimizer is adopted whose parameters β_1 and β_2 are set as 0.9 and 0.999. The learning rate is initialized to 1×10^{-4} and decreased following the cosine annealing strategy. The intra-frame model and the inter-frame model converge after 500000 and 700000 iterations, respectively. All experiments are conducted using the PyTorch with NVIDIA GTX 3090 GPUs.

3) *Test setting*: In order to compare rate-distortion performance with more video coding methods such as the commercial video codec x265[12], we generate the bitstream of x265 with FFmpeg. As mentioned above, to make comparison in more practical scenarios, we provide the RD results of x265, HM and VTM when intra period is set to 32. The detailed configurations of x265 veryslow is listed as follows:

- **x265 (LDP veryslow)**:
`ffmpeg -pix_fmt yuv420p -s Width×Height -r FrameRate
-i input.yuv -vframes Num -c:v libx265 -preset placebo`

TABLE I: BDBR(%) Results with the anchor of HM-16.22 random-access. **Red** and **Blue** indicate the best and the second-best performance.

Dataset	BDBR (%) calculated by Weighted YUV PSNR (6:1:1)		
	x265 veryslow	HM-16.22 LDP	BiTVC(Ours)
Validation1	87.49	27.15	-8.40
Validation2	157.14	36.43	-10.52
Validation3	199.81	54.60	19.20
Validation4	57.52	40.14	10.05
Validation5	106.16	22.50	16.99
Validation6	502.23	52.68	69.73
Validation7	64.73	12.42	46.69
Validation8	114.38	22.24	-26.48
Validation9	140.53	34.52	-11.61
Validation10	169.64	44.02	29.76

-tune zerolatency -x265-params
"crf=Q:keyint=IP" output.mkv

where the crf parameters (Q) are set to (19, 23, 27, 31), and intra period (IP) is set to 32. As for HM-16.22, we use the configurations as follows:

- **HM:**

Encode:

TAppEncoderStatic -c encoder_lowdelay_P_main.cfg or
encoder_randomaccess_main.cfg
-InputFile=input.yuv -InputBitDepth=8 -
InputChromaFormat=420 -FrameRate=FrameRate
-SourceWidth=Width -SourceHeight=Height -q QP -b
enc_output.bin -ip intraperiod -o enc_output.yuv

Decode:

TAppDecoderStatic -b dec_input.bin -o dec_output.yuv

B. Comparison with Other Methods

Table-I shows the experimental results on the Validation sequences while taking weighted YUV PSNR as quality measurements. When the intra period is set to 32, BiTVC achieves better rate-distortion performance than the x265 veryslow preset. Besides, the performance of our method can surpass HM-16.22 Low-delay P on all sequences. As a fully end-to-end optimized framework, our method exceeds HM random-access on four sequences: Validation1, Validation2, Validation8 and Validation9, which contain rich textual information and complex motion information.

C. Performance Analysis

To verify the effectiveness of the bidirectional prediction, we conduct experiments by changing unidirectional prediction to bidirectional prediction with the same bit rate model. In Figure 3, the blue curve denotes the baseline method LSTVC, and the orange curve represents BiTVC. We can observe that in the absence of the Bi-Prediction scheme and the

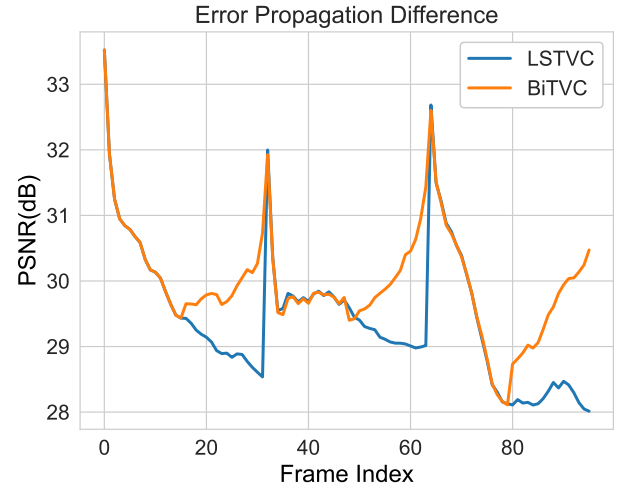


Fig. 3: Performance improvements from the Bi-Prediction and Bi Temporal Prior. The proposed scheme alleviates error propagation significantly compared to the baseline LSTVC.

TABLE II: The inference time cost per frame of HM-RA and BiTVC.

		High Bit Rate	Medium Bit Rate	Low Bit Rate
HM-RA	Encoding	17.91s	16.77s	15.94s
	Decoding	0.11s	0.07s	0.06s
BiTVC	Encoding	1.73s	1.74s	1.74s
	Decoding	0.57s	0.57s	0.57s

Bi Temporal prior, the reconstruction quality will decrease dramatically, especially in the large intra period situation.

1) *Model complexity and running time:* The parameters of the BiTVC are 49 megabytes. As for running time comparison, we take the sequences in the Validation dataset (1080P) to test inference time. We record the inference time of coding, including network forward and the entropy model. Even though auto-regressive prior can further boost performance, we do not choose it since it will slow down inference speed dramatically. It can be concluded that our proposed methods can achieve highly efficient compression.

IV. CONCLUSION

This paper presents a learned video compression framework that focuses on exploring diverse temporal information. The proposed methods BiTVC can utilize long-range, short-range, and bidirectional temporal information to mine the unique characteristic of video content. Specifically, for long- and short-range temporal information exploitation, we follow our previous work and adopt the temporal prior and the progressive guided motion compensation. Moreover, we extend the framework with bidirectional prediction to effectively alleviate error propagation. The three modules are complementary to each other and help to improve the framework significantly. When comparing in practical scenarios and setting intra period to 32, our method can also achieve comparable performance with the traditional standard reference software HM-16.22.

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